

3.2.6 Reactions of ions in aqueous solution (A-level only)

The reactions of transition metal ions in aqueous solution provide a practical opportunity for students to show and to understand how transition metal ions can be identified by test-tube reactions in the laboratory.

Content	Opportunities for skills development
In aqueous solution, the following metal-aqua ions are formed: $[M(H_2O)_6]^{2+}$, limited to $M = Fe$ and Cu $[M(H_2O)_6]^{3+}$, limited to $M = Al$ and Fe The acidity of $[M(H_2O)_6]^{3+}$ is greater than that of $[M(H_2O)_6]^{2+}$ Some metal hydroxides show amphoteric character by dissolving in both acids and bases (eg hydroxides of Al^{3+}). Students should be able to: - explain, in terms of the charge/size ratio of the metal ion, why the acidity of $[M(H_2O)_6]^{3+}$ is greater than that of $[M(H_2O)_6]^{2+}$ - describe and explain the simple test-tube reactions of: $M^{2+}(aq)$ ions, limited to $M = Fe$ and Cu, and of $M^{3+}(aq)$ ions, limited to $M = Al$ and Fe, with the bases OH^-, NH_3 and CO_3^{2-}	AT d and K PS 1.2 Students could carry out test-tube reactions of metal-aqua ions with $NaOH$, NH_3 and Na_2CO_3 AT d and k PS 2.2 Students could carry out test-tube reactions to identify the positive and negative ions in this specification. PS 1.1 Students could identify unknown substances using reagents.
Required practical 11 Carry out simple test-tube reactions to identify transition metal ions in aqueous solution.	